

Interlocking Elastic Piece

Some disclaimers

1. I can't get the pieces to separate, because the step file has them as one piece. Would need the step files for each piece, converted into solid parts, then a solid assembly
2. Because the pieces aren't separate bodies, I also can't get cross cut sections, which is a shame because that's where most of the stress/strain expresses itself
3. Checkout the Study Reports for cool pics/numbers

Torsion Test

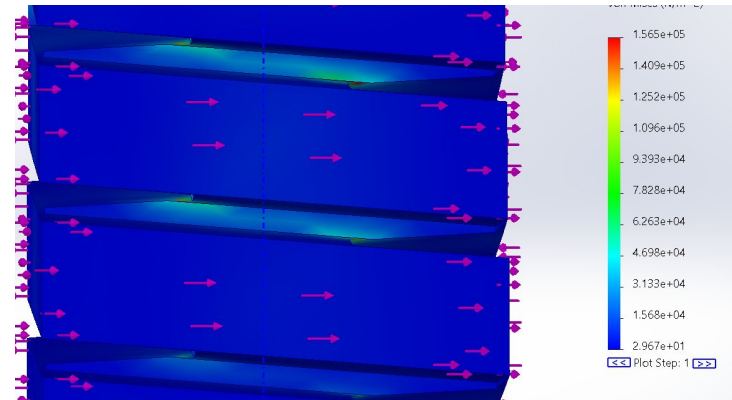
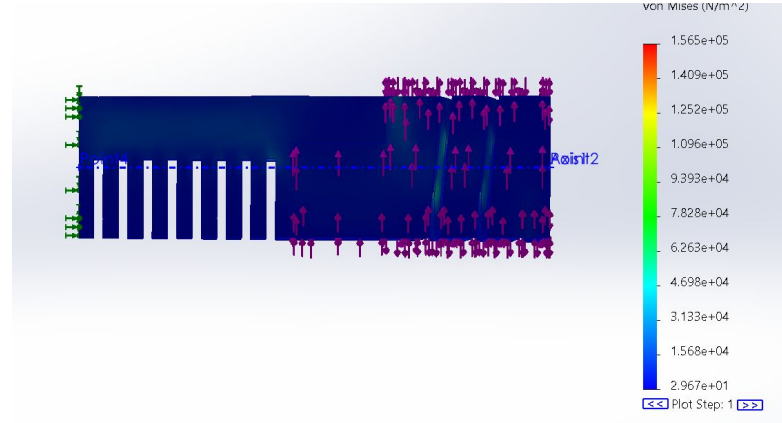
The setup:

- The piece on the left is “fixed”, i.e. the left hand is holding the piece on the left
- The piece on the right is being twisted away from the body (most people are right handed)
- The torque exerted is $\sim 1\text{Nm}$
- The average torque for what people use for a screwdriver for “hand tight” is $\sim 5\text{ Nm}$, max, so 1 Nm is pretty little
 - Source: The NASA Anthropometric Source Book (enthusiastically cited in ergonomics and engineering forums) uses muscle strength data from isometric studies (such as Asmussen & Heeboll-Nielsen 1961, Backlund & Nordgren 1968, Lauback 1976) to support these torque ranges, confirming the “hand tight” value is around 4–5 Nm for average adults

Torsion Test

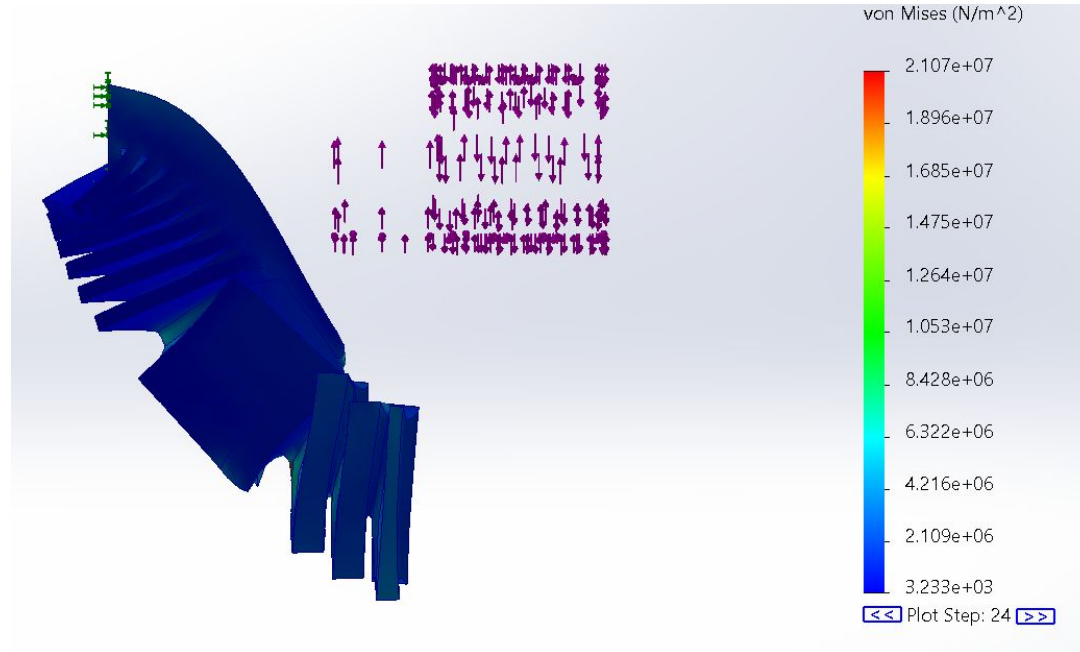
Initial State: Stress

- It's a little bit hard to see, but the stress is on the inside portion of this spiral piece



Torsion Test

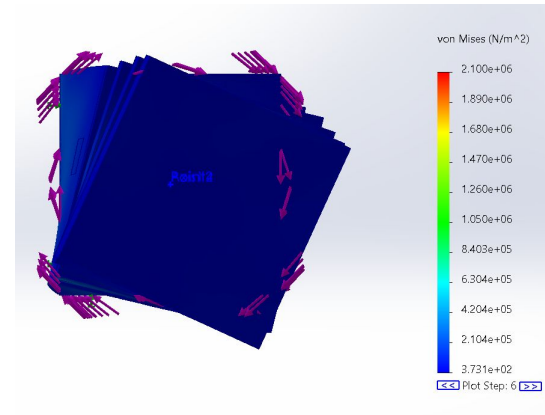
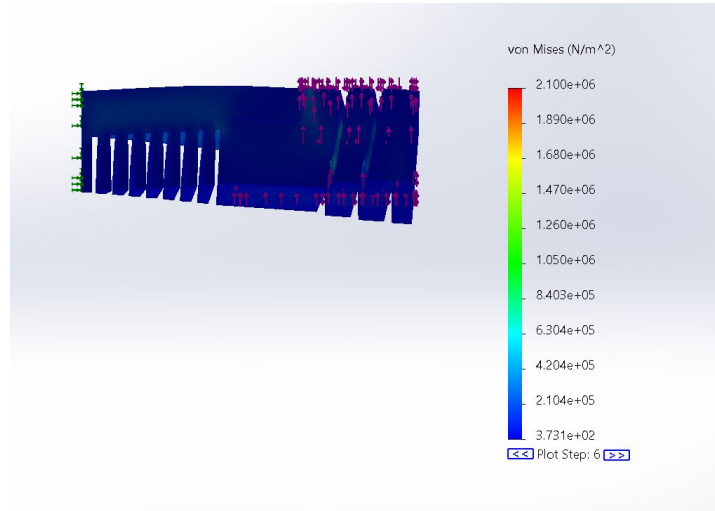
Final State: Stress



Torsion Test

Intermediate State: Stress

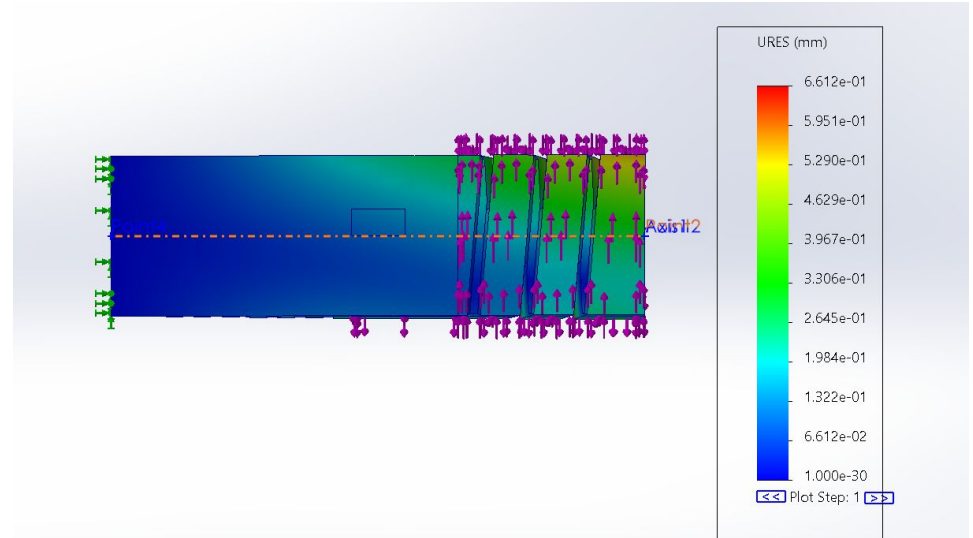
- If i had to guess where the two pieces would pop apart, it would be here
- I included a profile shot, so you can better see the torsion



Torsion Test

Initial State: Deformation

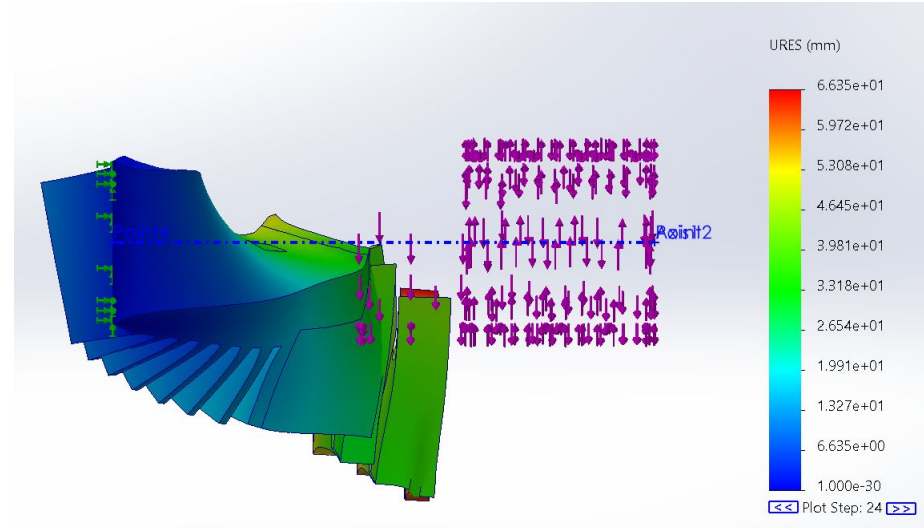
- You can see the left hand holding, right hand twisting a lot better in this one



Torsion Test

Final State: Deformation

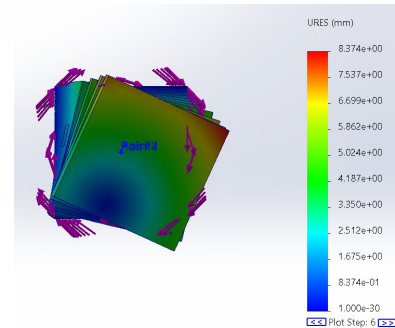
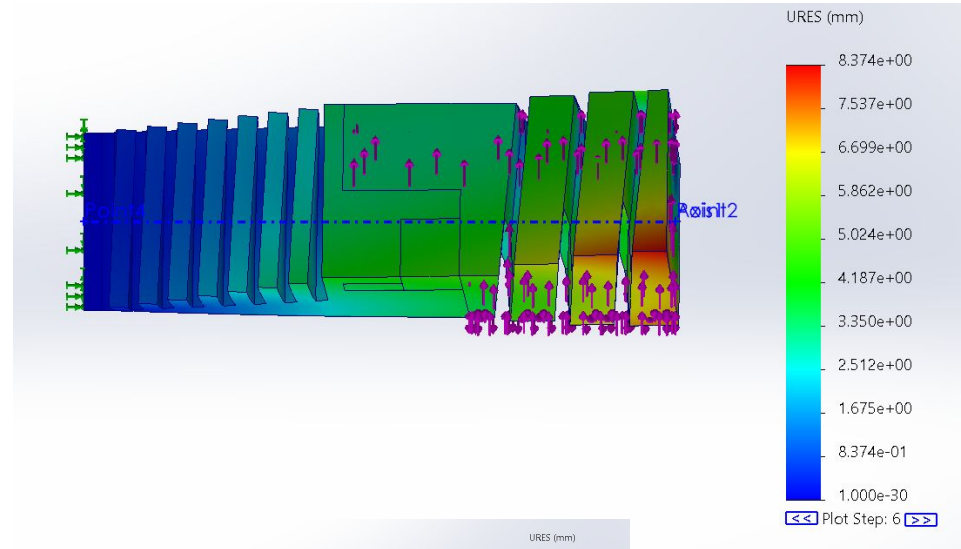
- Mm flexible
- I don't think result is real



Torsion Test

Intermediate State: Deformation

- This is probably more realistic to where the contacts would pop off
- I included a side profile here
 - The blue square in the back is the left hand holding
 - The green/yellow/red/blue square we are looking at is the left hand twisting
 - When not twisted, the corners of the squares are aligned, meaning 0 deformation
 - When twisted, the left hand (the face that we are looking at) is displaced ~45 degrees off axis from the original state (the blue/right hand square) which is why I think that it would pop off
 - If it were twisted so that the squares were 90 degree offset (square for right hand, diamond for left hand), it would mean that we have to compress the material and this material has a Poisson ratio of 0.49, which is nearly incompressible
 - **TL;dr – you would have to compress the material beyond this and that's not possible**

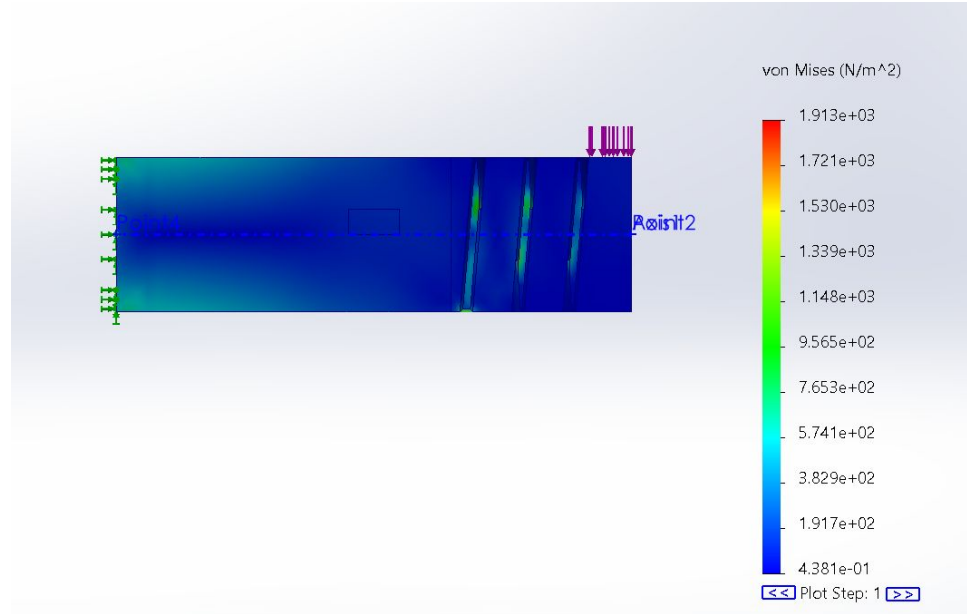


Bending Test

- Force to push down: 0.5N ~approximately the force used to press a key on a keyboard
- Source: “To accommodate a mean relaxed finger weight, a .50 (50.6 grams) Newton key activation force would be required.”
[https://www.resna.org/sites/default/files/legacy/conference/proceedings/2004/Papers/StudentDesign/OUT/KeyboardForce.html#:~:text=Another%20study%20at%20the%20University,47.6%20grams\)%20\(4\).](https://www.resna.org/sites/default/files/legacy/conference/proceedings/2004/Papers/StudentDesign/OUT/KeyboardForce.html#:~:text=Another%20study%20at%20the%20University,47.6%20grams)%20(4).)
- Same thing here, the right hand is still, the left hand/finger is pushing down

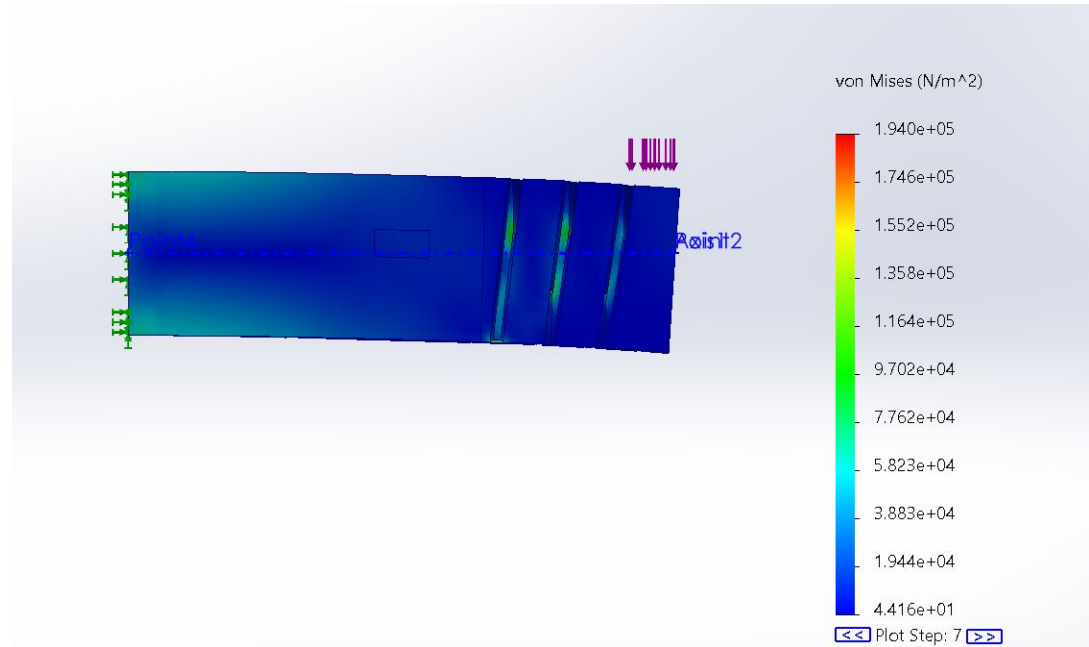
Bending Test

- Initial State: Stress



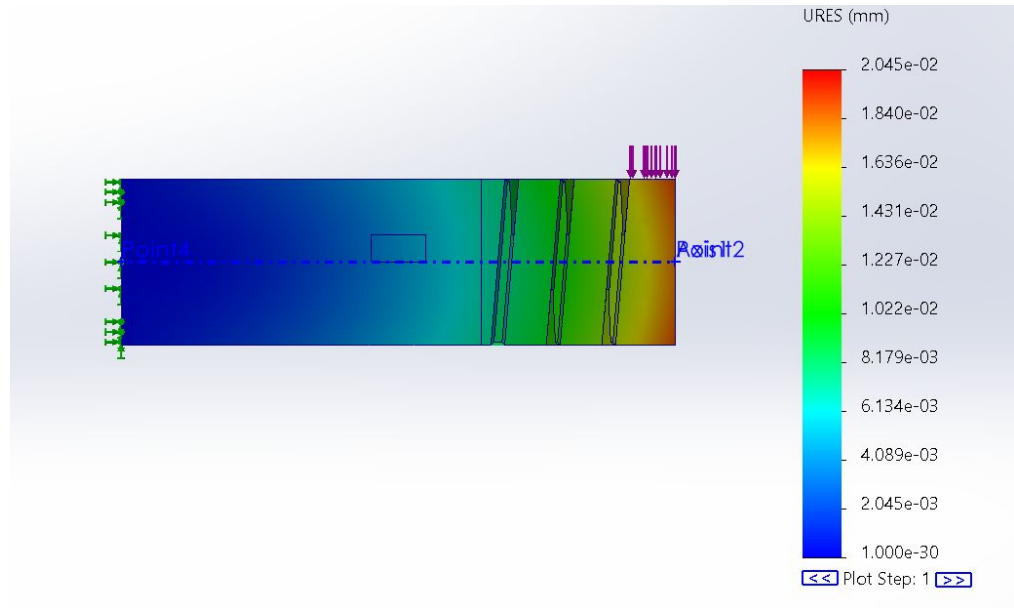
Bending Test

- Final State: Stress
- Wow, so exciting



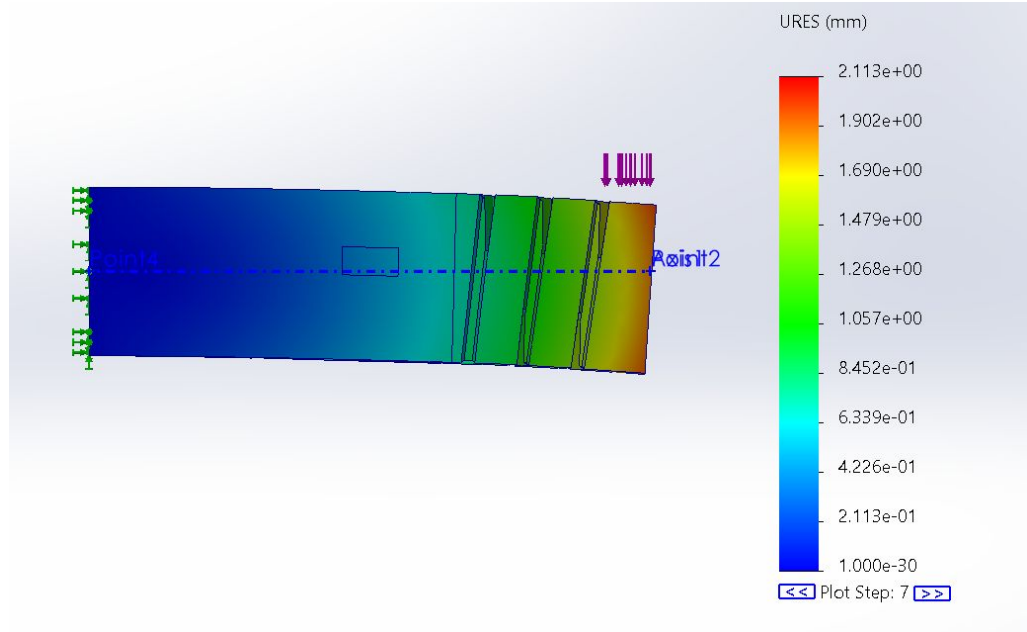
Bending Test

- Initial State:
Displacement
- Literally a colorful
fbd



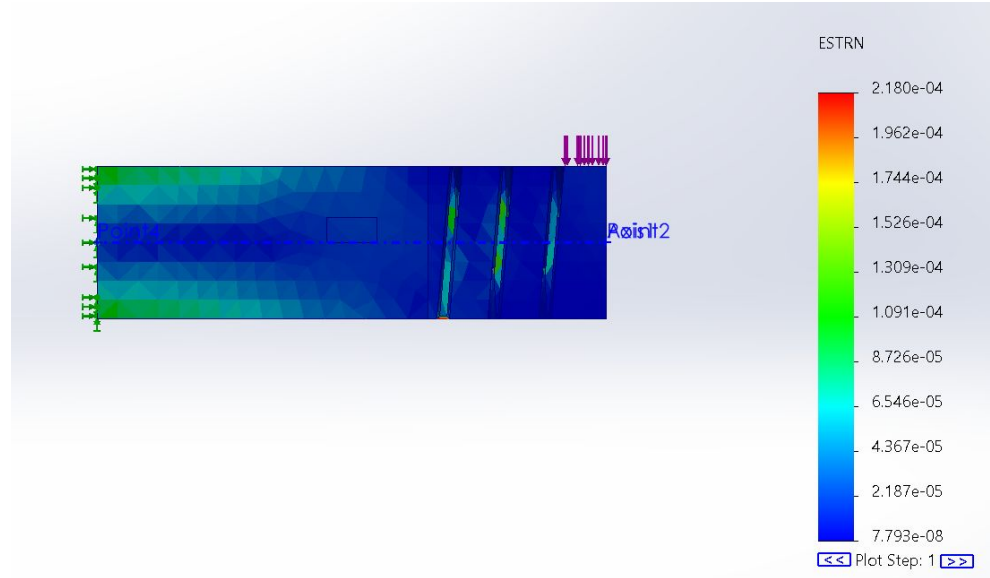
Bending Test

- Final State:
Displacement
-



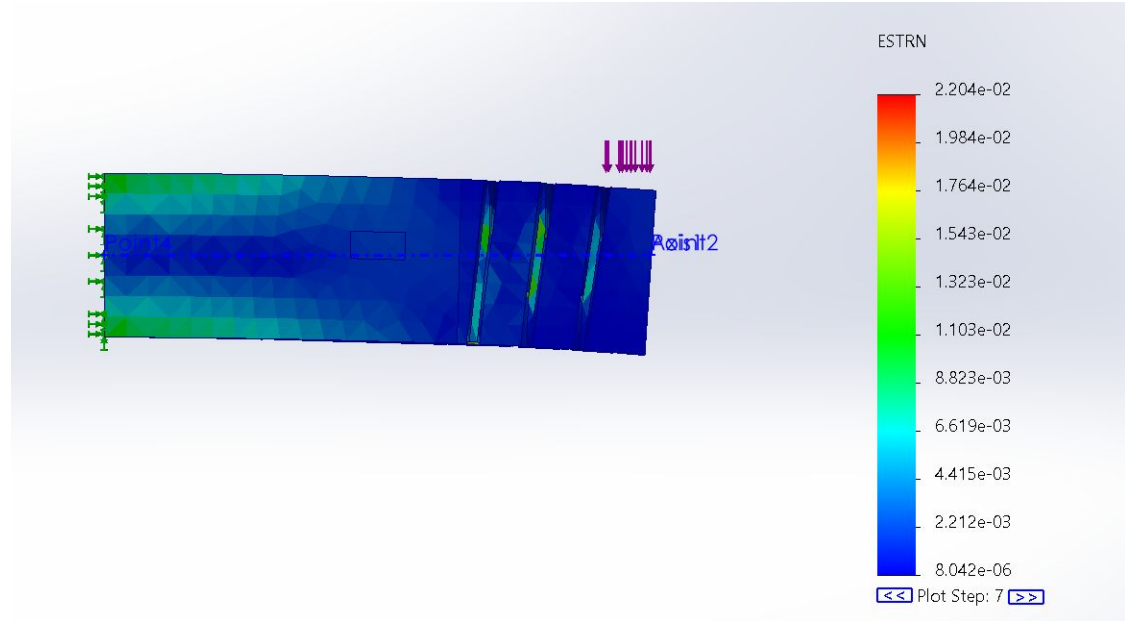
Bending Test

- Initial State: Strain
- You can see what is supporting the piece in green



Bending Test

- Final State: Strain
- You can see what is supporting the piece in green

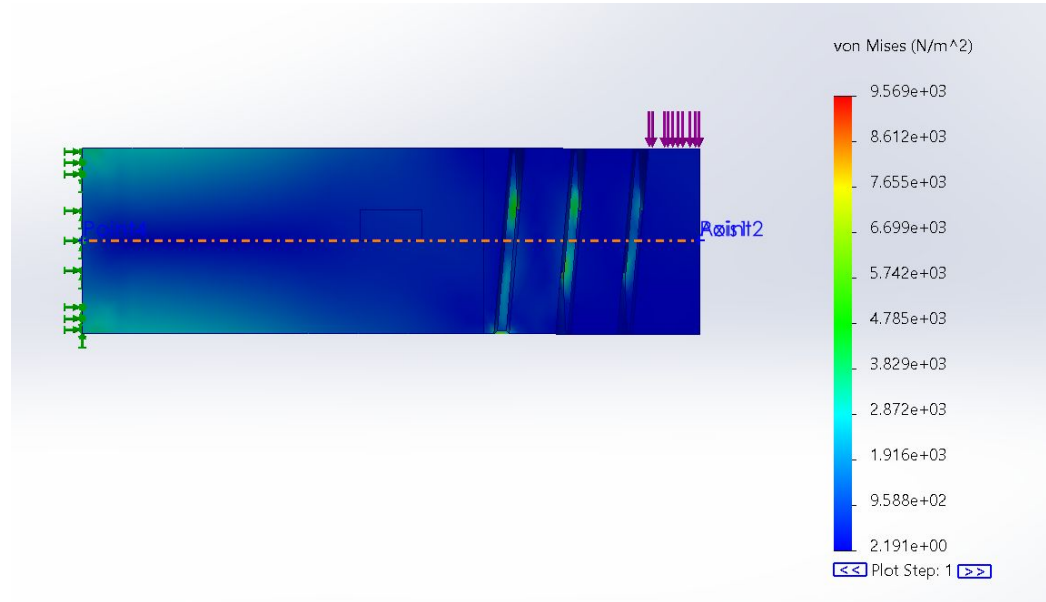


Bending Study 2: More force

- Push down with $\sim 2.5\text{N}$
- Source: “Another study at the University of Michigan determined that the overall mean peak reaction force during a typing task is 2.54 Newtons (263.1 grams)”
- Same link as before
-

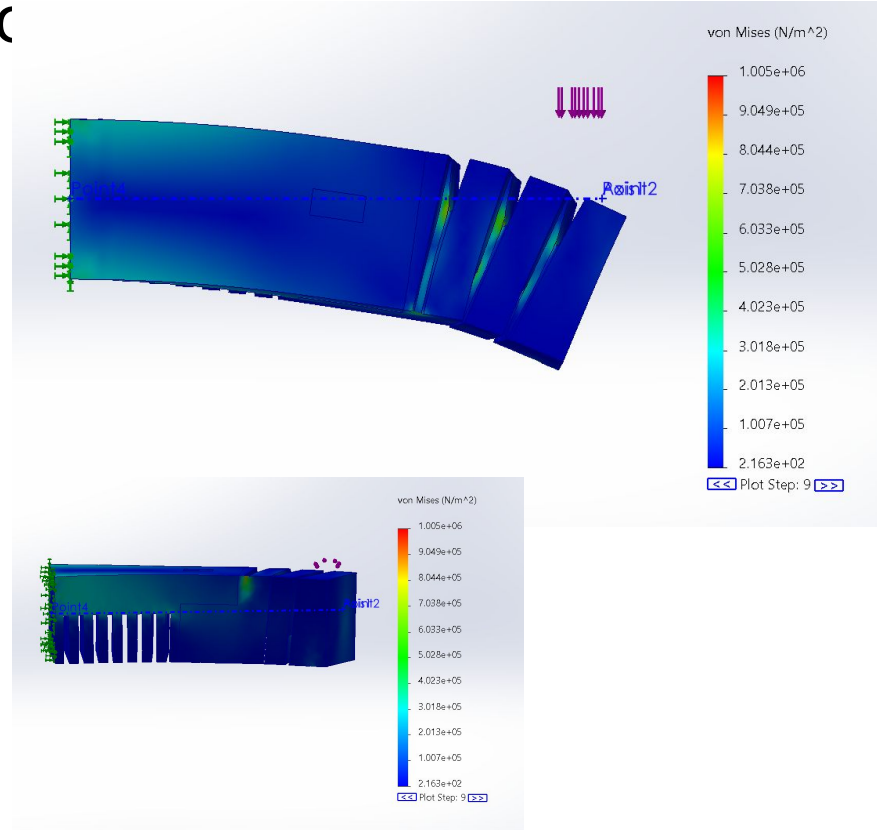
Bending Study 2: More force

- Initial State: Stress
-



Bending Study 2: More for

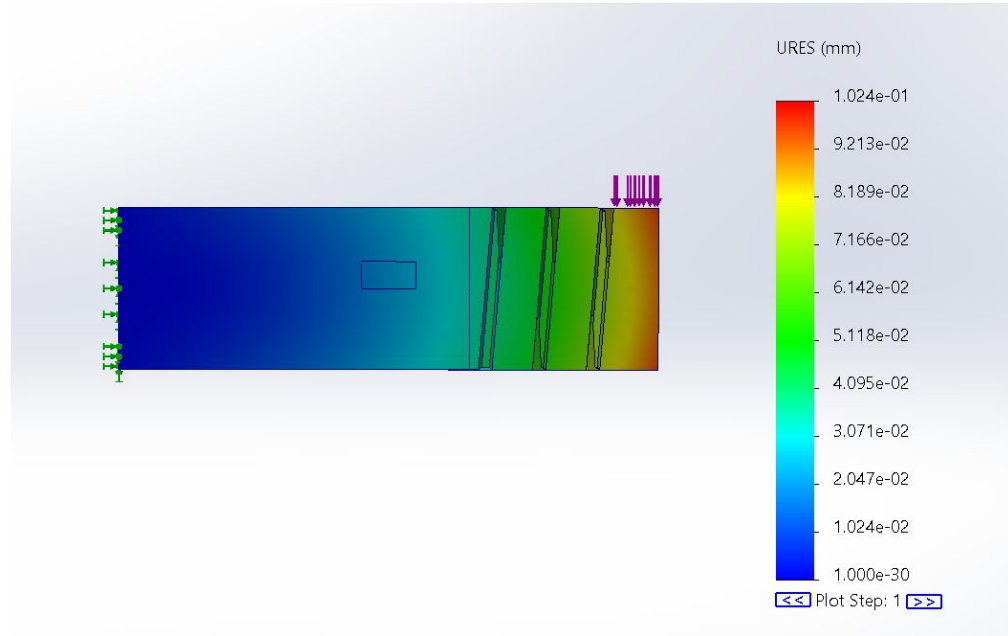
- Final State: Stress
- Included a shot of the underside, where the fins are



Bending Study 2: More force

- Initial State:
Displacement

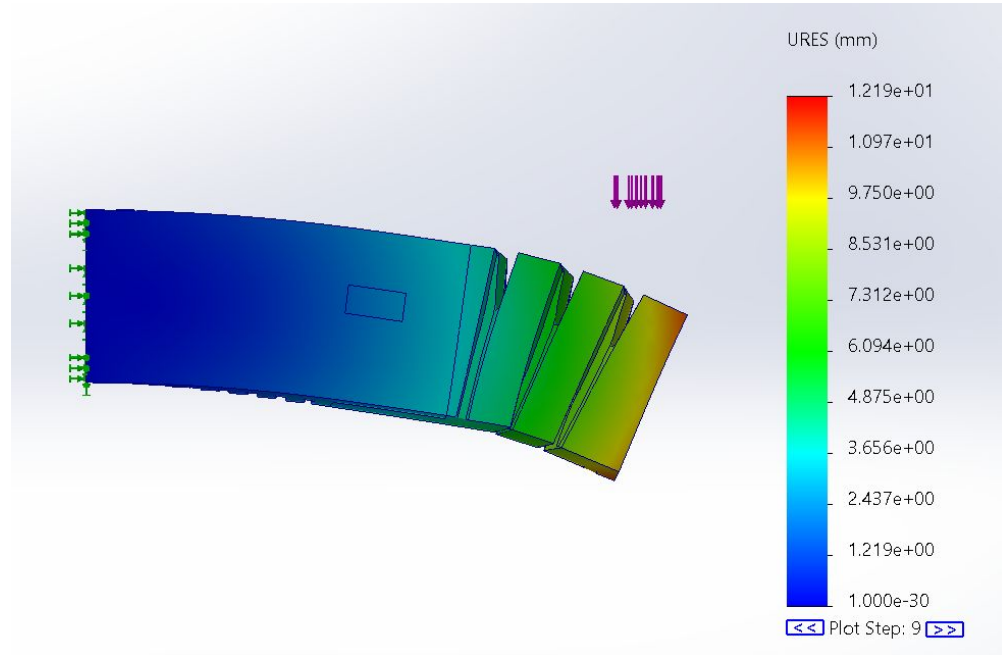
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Bending Study 2: More force

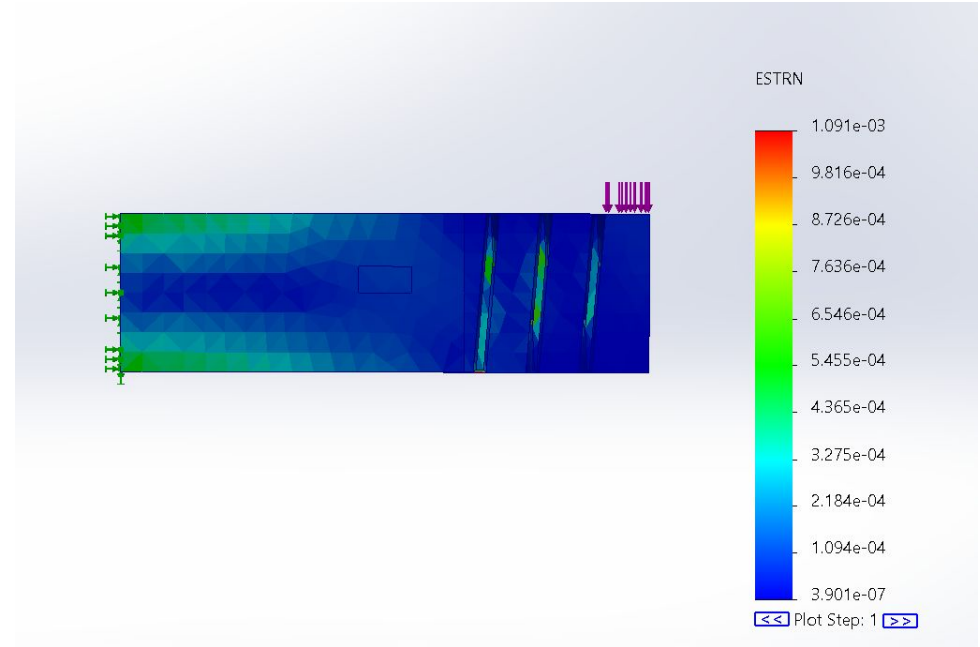
- Final State:
Displacement

-



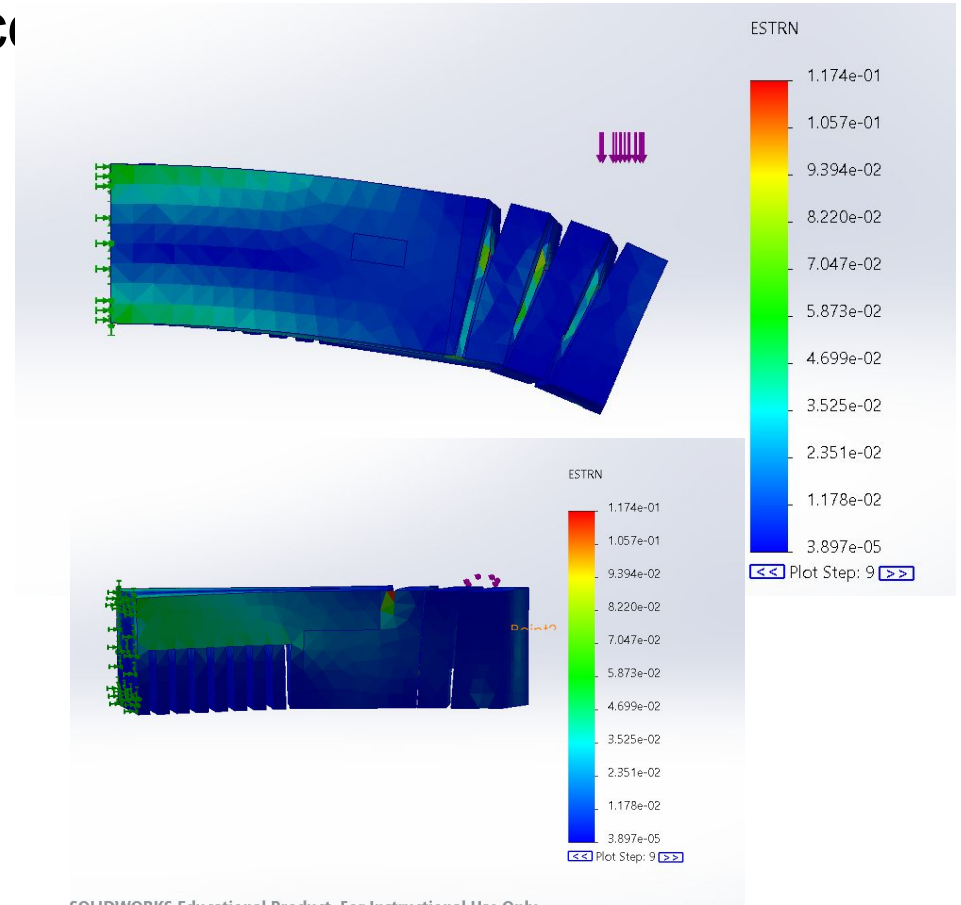
Bending Study 2: More force

- Initial State: Strain
-



Bending Study 2: More force

- Final State: Strain
- Plus shot of the underside

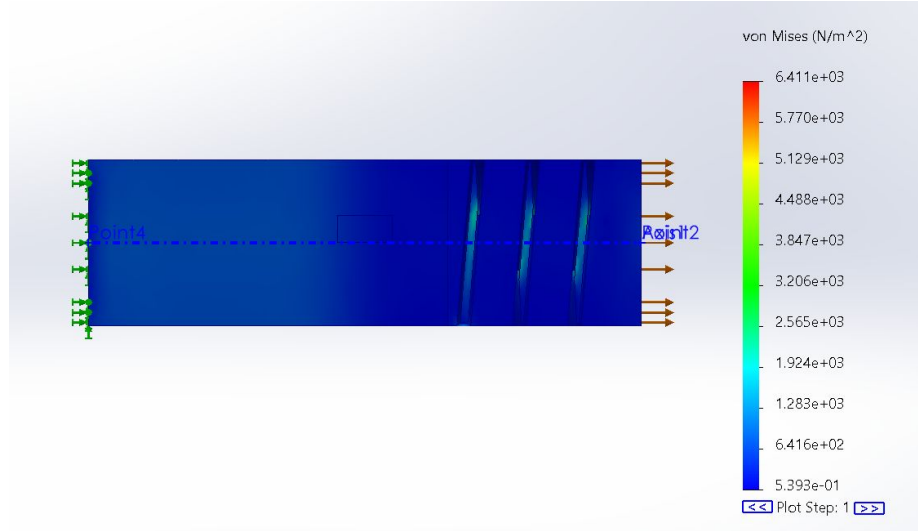


Tension Study

- How much force required to pull a door open: ~22N
- Source:
<https://idighardware.com/2009/02/door-opening-force/#:~:text=Here%20are%20some%20maximum%20forces%20for%20opening,hold%20the%20door%20in%20a%20closed%20position>
-
- That seems like a lot, so arbitrary value would be 5N, which is the force required for pushing an empty/mostly empty shopping cart

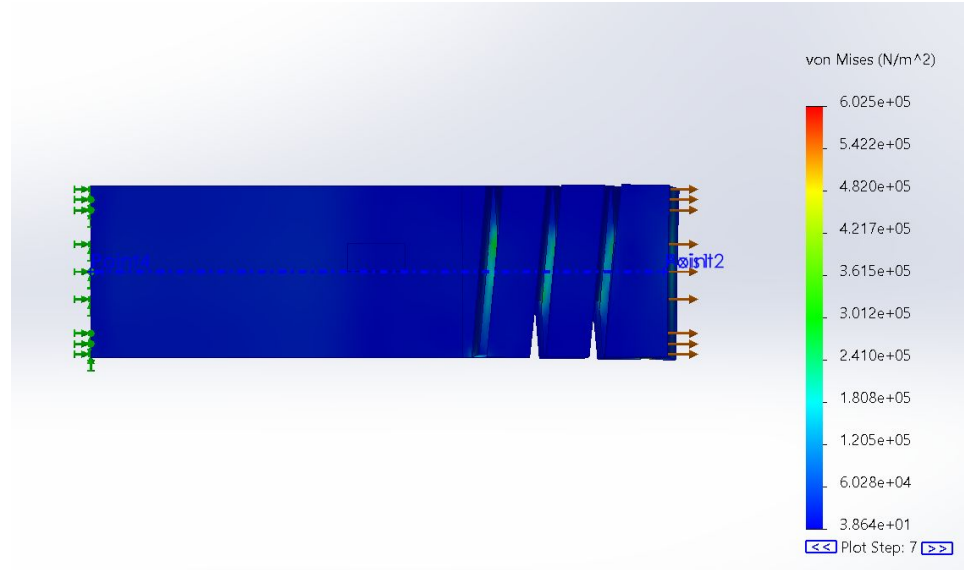
Tension Study

- Initial State: Stress



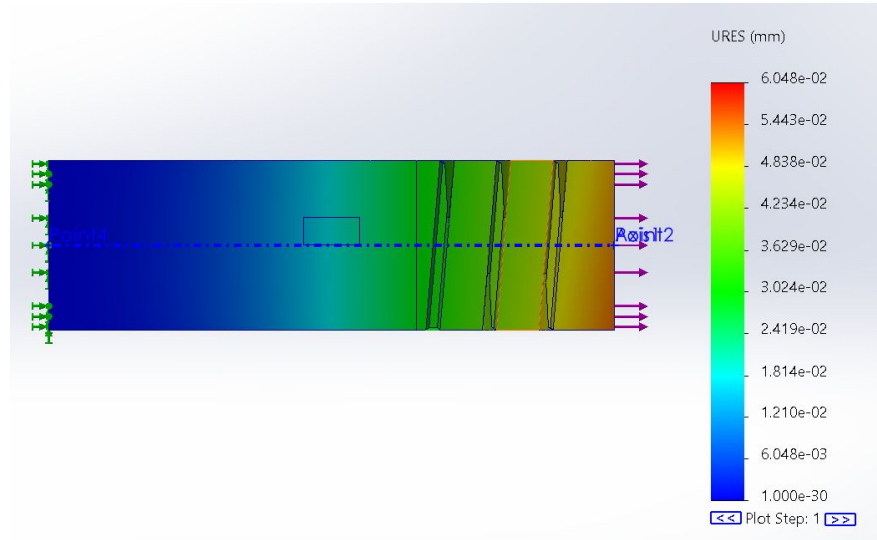
Tension Study

- Final State: Stress



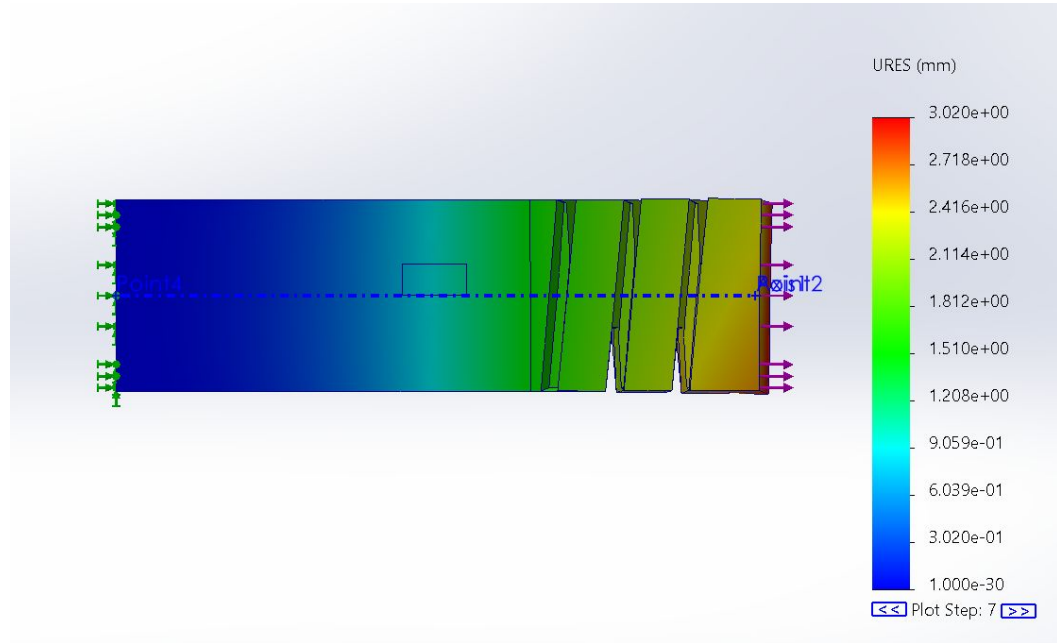
Tension Study

- Initial State:
Displacement



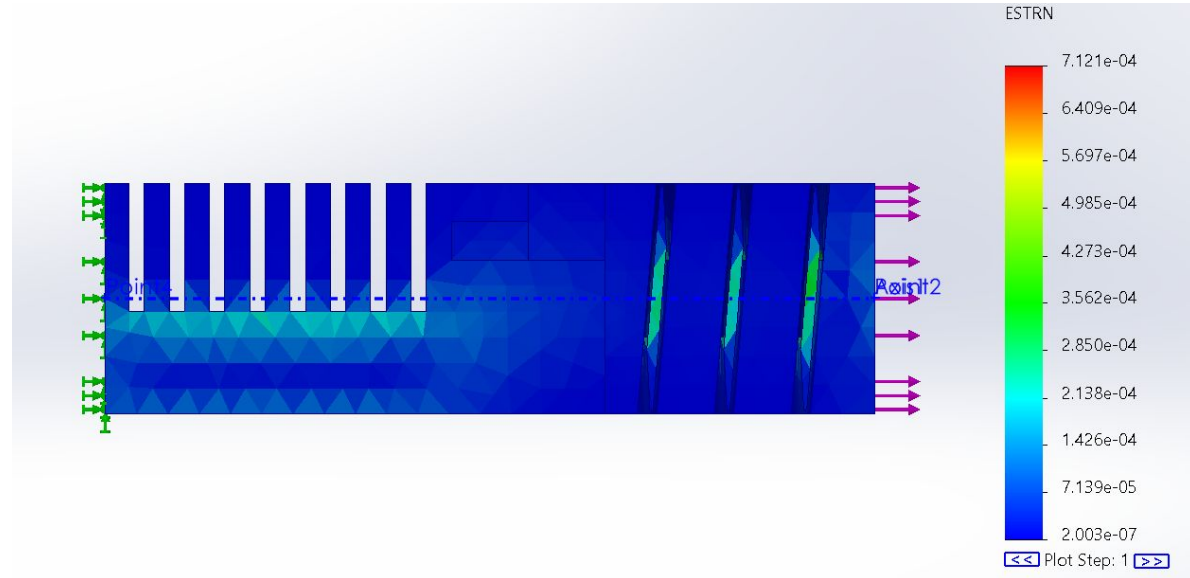
Tension Study

- Final State:
Displacement



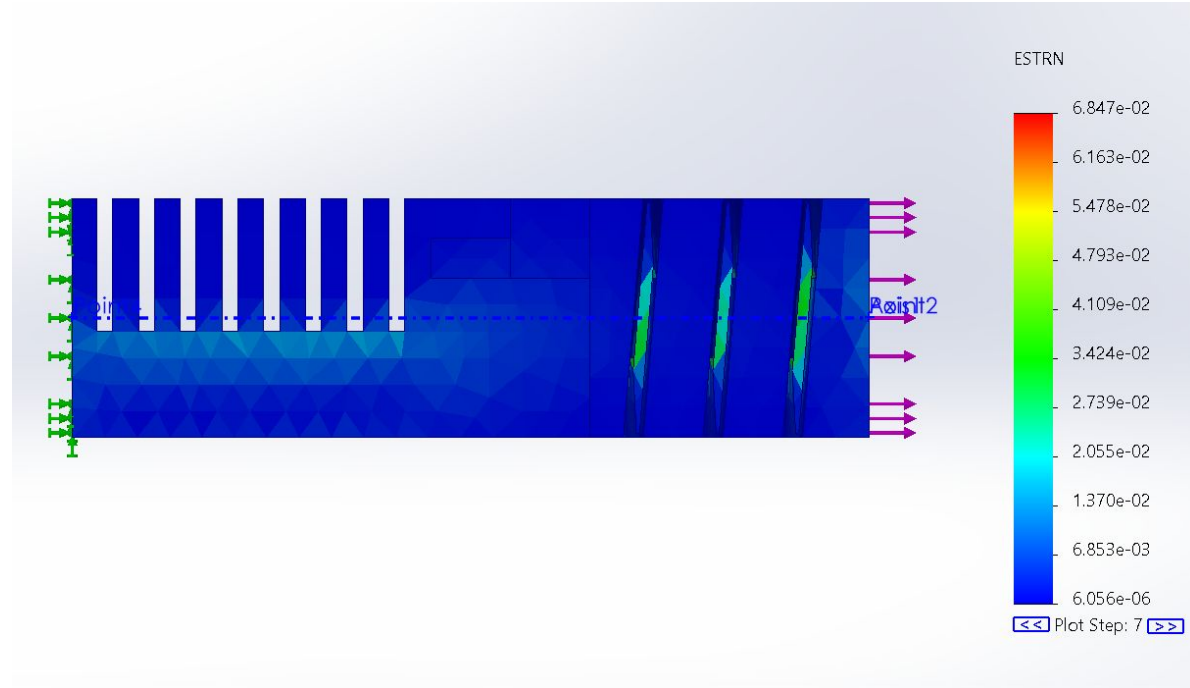
Tension Study

- Initial State: Strain



Tension Study

- Final State: Strain

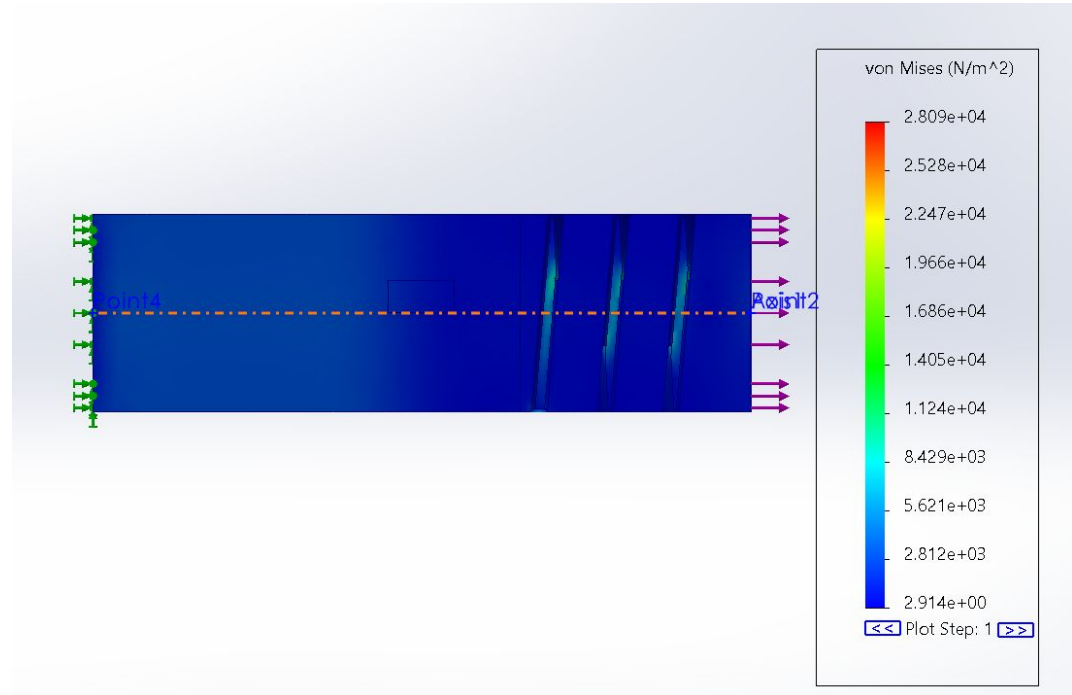


Tension Study 2: Full Strength

- Force at 22N

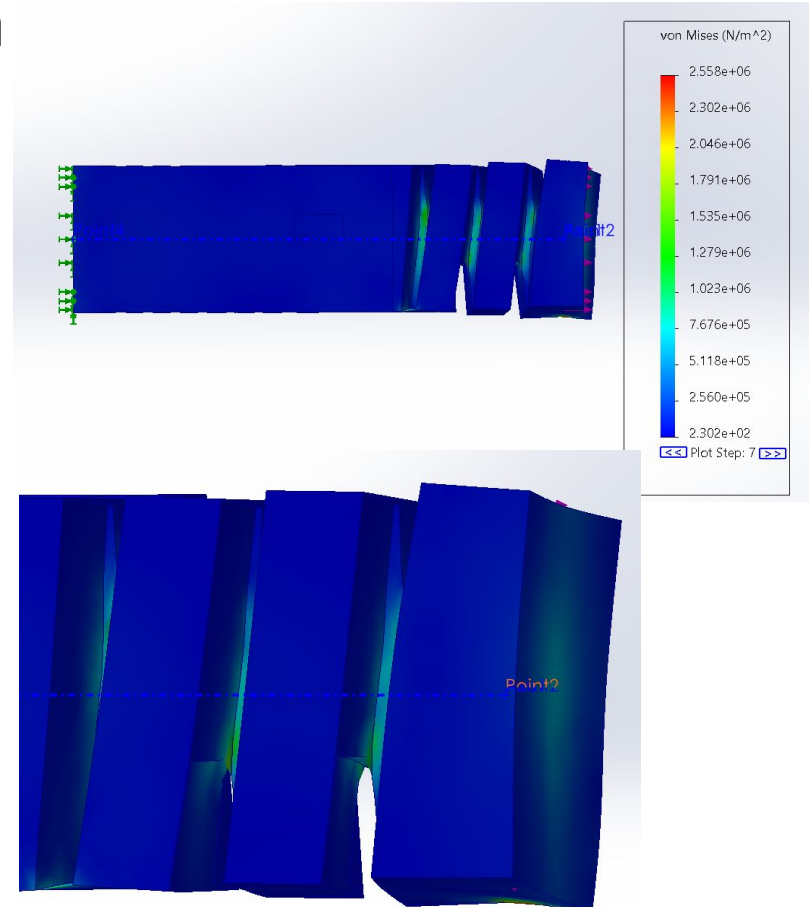
Tension Study 2: Full Strength

- Initial State: Stress



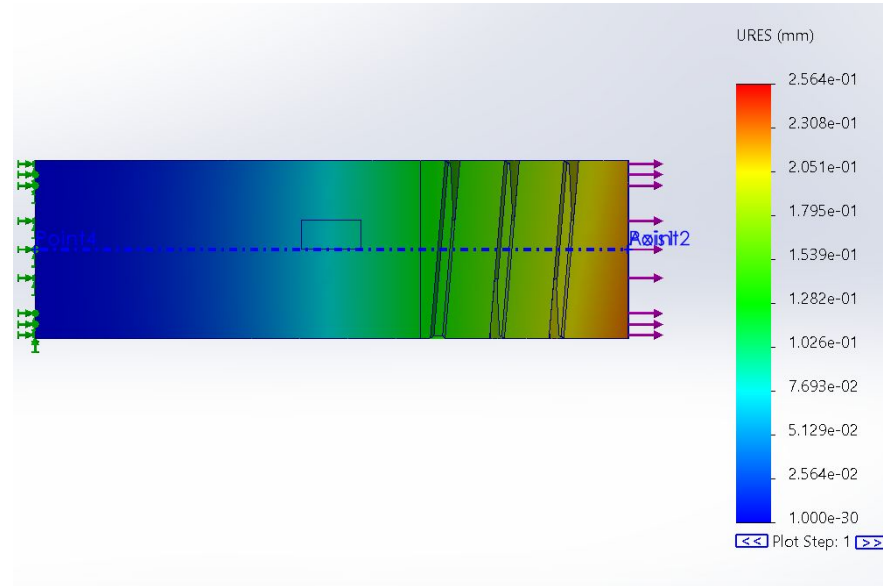
Tension Study 2: Full Strength

- Final State: Stress
- This is kinda cool, you can see the edges of the left side flare out bc they're free and the middle is where it is connected to the spiral/rest of the piece so it is sunken in a little bit



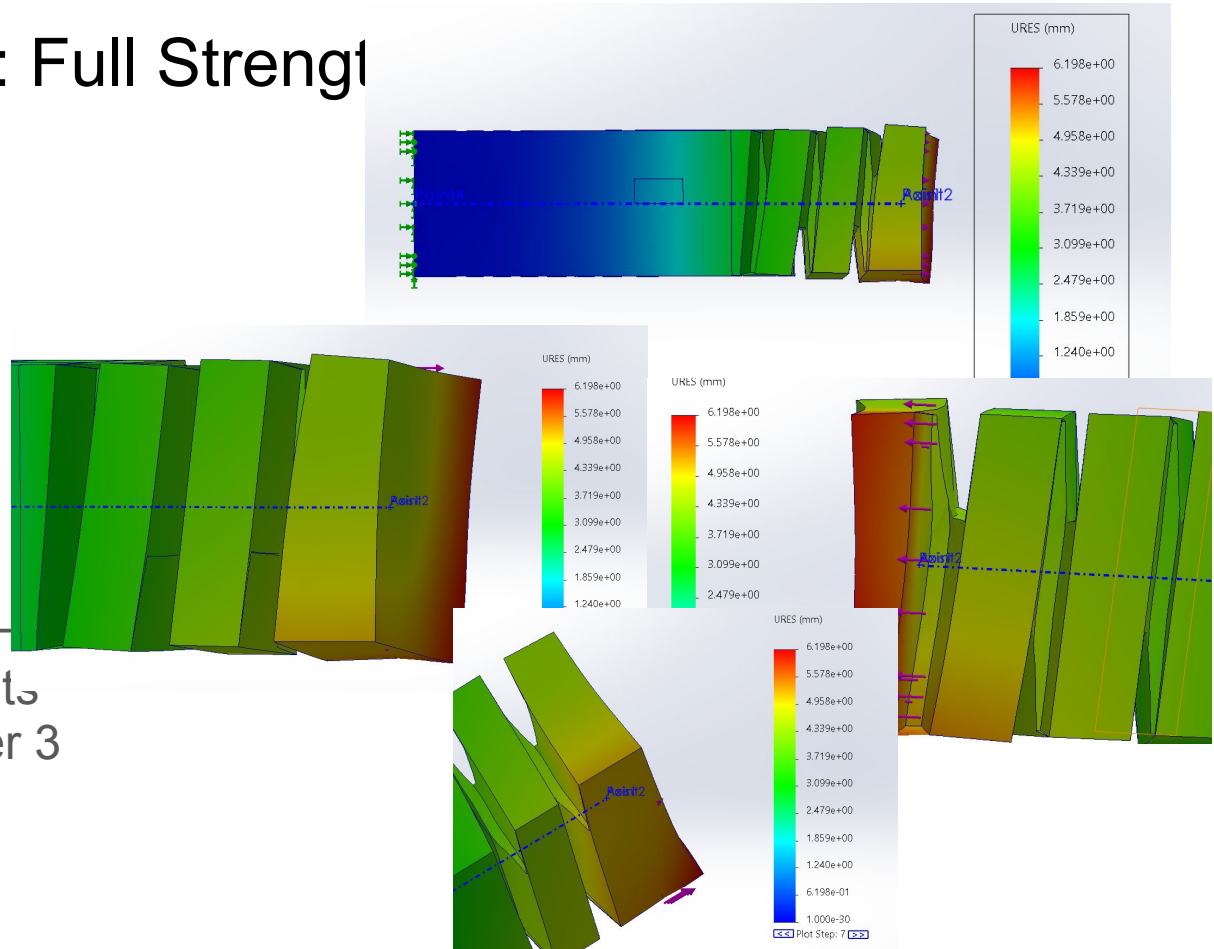
Tension Study 2: Full Strength

- Initial State:
Displacement



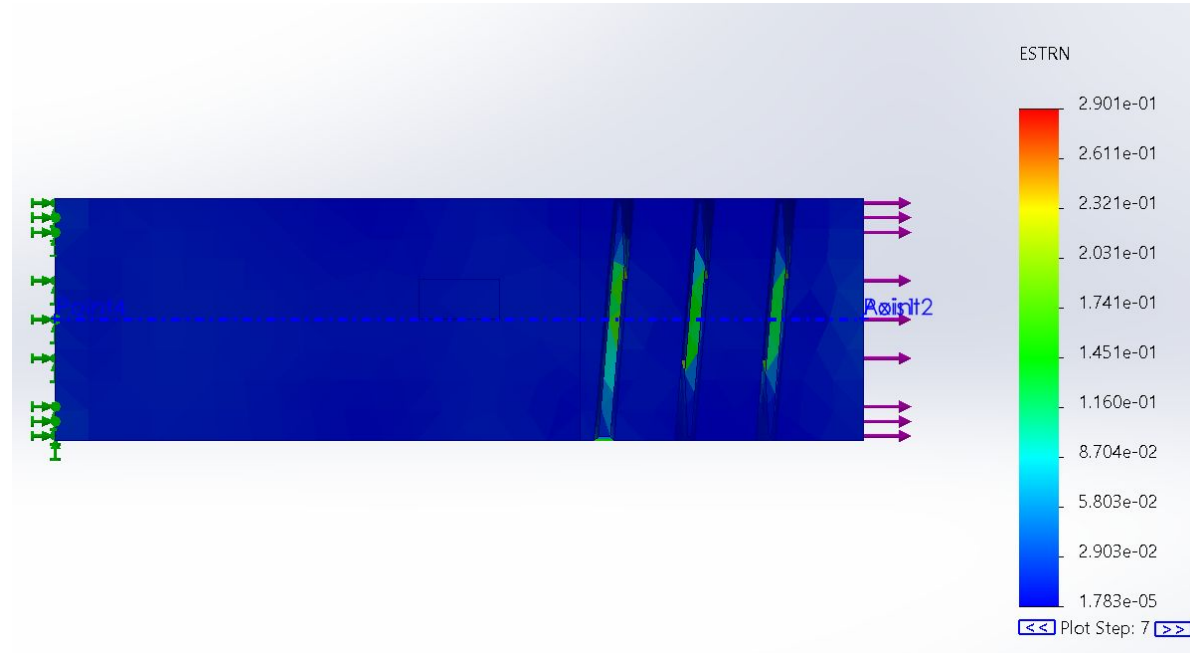
Tension Study 2: Full Strength

- Final State:
Displacement
- The red edge is thinner and less supported so it flexes more
- You can see it in comparison to the other edges OF THE SAME FACE that it's thicker on the other 3 edges



Tension Study 2: Full Strength

- Initial State: Strain



Tension Study 2: Full Strength

- Final State: Strain

